

Year 9 Semester Two Test

Allowed: A4 page of notes and calculators
Time: 55mins

Name: _____

1. What is the probability of getting less than 5 when rolling a dice?

- a) $\frac{1}{6}$ b) 67% c) 0.4 d) 73% e) 0.8

[1 mark]

2. Pond has a surface area of $4m^2$ and a depth of 2m. What is the volume?

- a) $6m^3$ b) $2m^3$ c) $8m^3$ d) $24m^3$ e) $12m^3$

[1 mark]

3. The Tangent ratio is found using which of the following?

- a) Opposite $\div$ Hypotenuse b) Adjacent $\div$ Hypotenuse c) Adjacent $\div$ Opposite

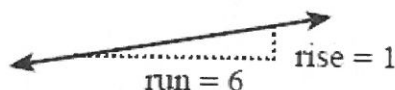
[1 mark]

- d) Hypotenuse $\div$ Opposite e) Opposite $\div$ Adjacent

4. Which of the following is 0.0056 written correctly in Scientific Notation?

- a) 5.6×1000^{-1} b) 56×10^{-3} c) $5.6 \div 10^{-3}$ d) $5.6 \div 1000^1$ e) $5.6^3 \times 10$

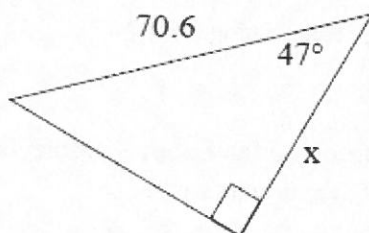
[1 mark]



5. What is the gradient of the line above?

- a) $\frac{1}{6}$ b) 6 c) 16 d) 61 e) 6 f) 5

[1 mark]



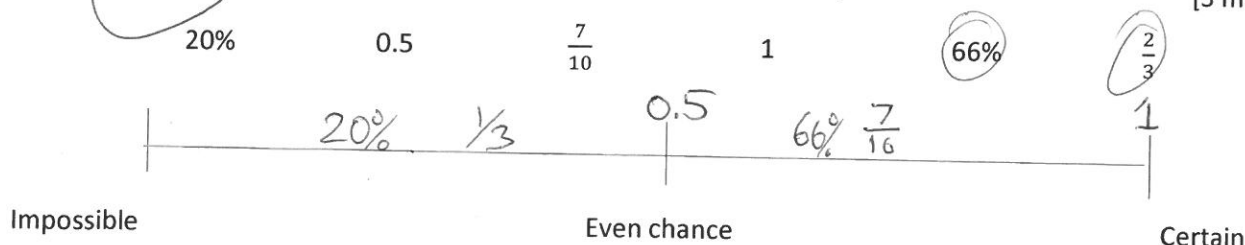
6. What is the name of the side marked with an "x" in the triangle above?

- a) Adjacent b) Opposite c) Hypothetical d) Tangent e) Hypothesis

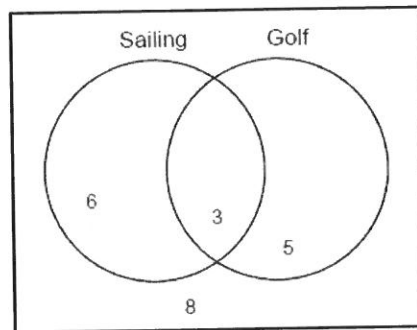
[1 mark]

7. Draw a line to place the following numbers onto the probability scale.

[3 marks]



8. Use the Venn diagram to answer the following questions?



a) How many students like sailing?

[1 mark]

9

b) How many students do not like golf?

[1 mark]

14

c) What is the probability that a student picked at random, does not like either sailing or golf?

[1 mark]

$$P = \frac{8}{22} = \frac{4}{11}$$

9. A recent survey of 1000 car owners is summarised in the table below.

Car type	Frequency
Sedan	212
Hatchback	332
Van	189
Utility	253
Convertible	14

a) What is the relative frequency of owning a sedan? *(as a decimal / percentage)*

[1 mark]

$$\frac{212}{1000} = 0.212$$

b) What is the relative frequency of not owning a van?

[1 mark]

$$\frac{811}{1000} = 81\%$$

c) Using the relative frequencies from the table above, predict how many Utility drivers there would be if there were 75,000 drivers survey.

[2 mark]

$$0.253 \times 75\,000 = 18\,975$$

10. Expand the following.

[5 marks]

a. $4(x + 2) =$

$$4x + 8$$

b. $-3(2t + 2) = -6t - 6$

c. $2(1 - r) = 2 - 2r$

d. $(g + 1)(g + 4) =$

$$g^2 + 4g + g + 4 = g^2 + 5g + 4$$

e. $(2e + 3)(e - 4) =$

$$2e^2 - 8e + 3e - 12 = 2e^2 - 5e - 12$$

11. Factorise the following.

a. $6t + 2$

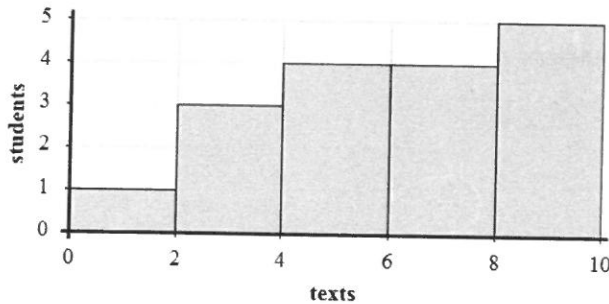
$$2(3t+1)$$

b. $12ag + 4a$

$$4a(3g+1)$$

[2 marks]

12. The histogram below shows the number of texts students sent each day.



[1 marks]

1) Most students sent between 2 and 10 texts.

2) How many students sent between 8 and 10 texts?

5

[1 marks]

3) How many students are represented in this histogram?

17

[1 marks]

4) If a student sent 2 texts which bar would they be added to?

1st

[1 marks]

13. The star Sirius is approximately 75 684 000 000 000 km from the Sun.

[1 mark]

a. Write this in scientific notation.

$$7.5684 \times 10^{13}$$

The Apollo moon landing rocket could travel at 3.5×10^8 km per year.

[1 mark]

b. Write this as a regular number.

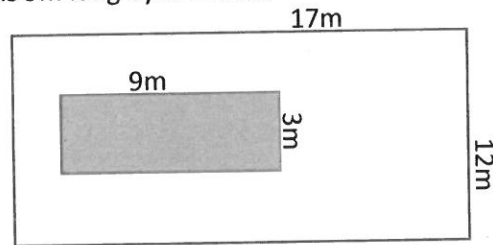
350 000 000

c. How long would it take to travel to Sirius from the Sun if you were to travel at that speed?

[2 mark]

$$\frac{7.5684 \times 10^{13}}{3.5 \times 10^8} = 216\,240 \text{ years}$$

14. Melissa is getting her rectangular back yard paved and at the same time she has also decided to get a pool. The backyard is 12m wide by 17m in length and the pool is also a rectangle that is 9m long by 3m wide.



- a. What is the size of the area that Melissa needs to get paved?

[1 mark]

$$A = (17 \times 12) - (9 \times 3)$$

$$= 204 - 27$$

$$= 177 \text{ m}^2$$

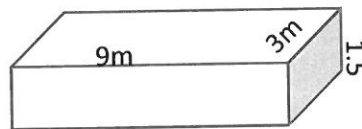
- b. If the pool is 1.5m deep (assume it is 1.5m deep from end to end), what is the volume of the pool?

[1 mark]

$$V = l \times w \times h$$

$$= 9 \times 3 \times 1.5$$

$$= 40.5 \text{ m}^3$$



- c. How many litres of water will be needed to fill the pool?

[1 mark]

$$40.5 \text{ m}^3 = 40.5 \text{ kL}$$

$$= 40\,500 \text{ L}$$

15. From the graph to the right;

- a. What are the coordinates of point A? [1 marks]

- b. What are the coordinates of point B? [1 marks]

(1, 3)

- c. What is the gradient of the line? [1 marks]

$$\frac{4}{6} = \frac{2}{3}$$

- d. How long is the line between points A and B? [1 marks]

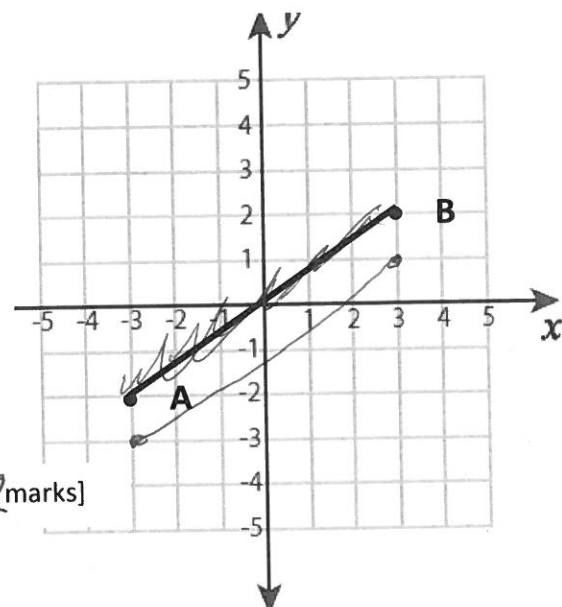
$$AB^2 = 4^2 + 6^2$$

$$AB = \sqrt{52}$$

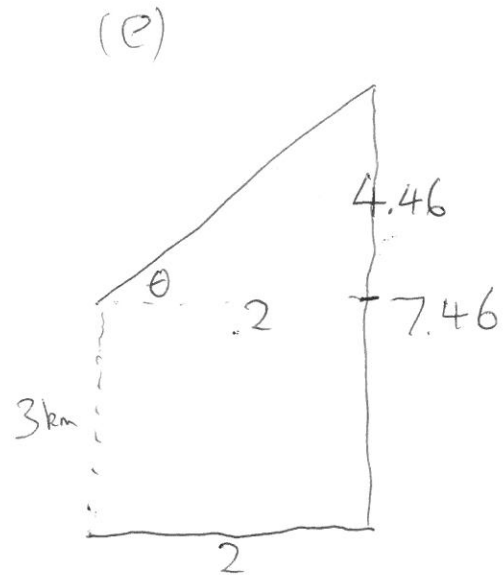
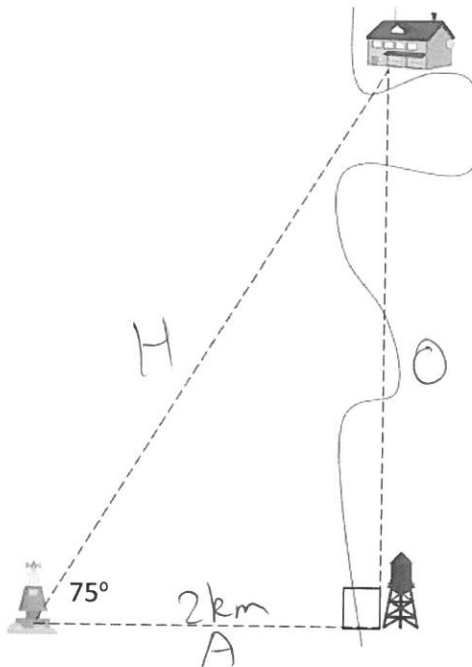
$$AB = 7.21$$

- e. What are the coordinates for the midpoint? [1 marks]

(0, -1)



16. John is sailing past a navigation marker toward his yacht club. The navigation marker is directly off the coast from a water tower (so the three objects form a right angled triangle as shown below).



- a) Label the sides of the triangle above using Opposite, Adjacent and Hypotenuse.

[1 marks]

John checks his chart and can see that there is 2km from the water tower to the navigation marker.

- b) What rule can John use to find how far he is from the yacht club?

[1 marks]

Tangent Cosine

- c) Find the distance John needs to travel to arrive at the Yacht club (to 2dp).

[2 marks]

$$\begin{aligned} \cos 75 &= \frac{2}{H} \\ H &= 2 \div \cos 75 \\ H &= 7.73 \text{ km} \end{aligned}$$

- d) Calculate the distance from the water tower to the yacht club (to 2dp).

[2 marks]

$$\begin{aligned} \tan 75 &= \frac{O}{2} \\ O &= 2 \times \tan 75 \\ O &= 7.46 \text{ km} \end{aligned}$$

- e) If John continues heading straight along the coast for 3km (so that he stays 2km offshore), what is the size of the angle formed between the yacht club, the boat and the shore (to the nearest degree)?

[2 marks]

$$\begin{aligned} 7.46 - 3 \\ = 4.46 \text{ km} \end{aligned}$$

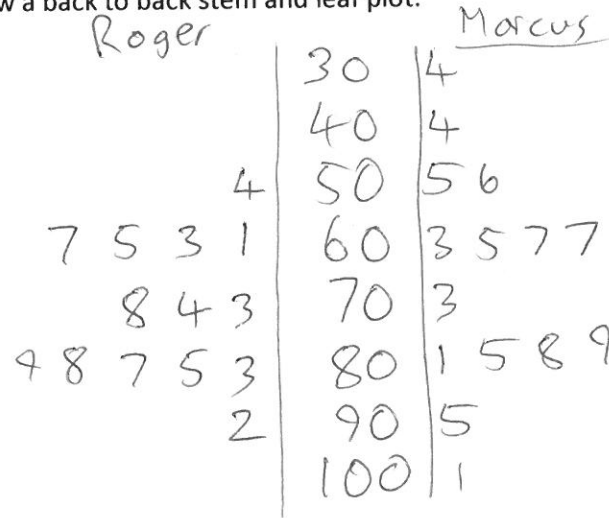
$$\begin{aligned} \tan \theta &= \frac{4.46}{2} \\ \theta &= 66^\circ \end{aligned}$$

17. Marcus and Roger were arguing whose basketball team was better. They looked at the high scores from there last season.

Marcus's team 55, 67, 44, 89, 67, 56, 88, 101, 95, 34, 65, 63, 85, 81, 73
 Roger's team 67, 74, 88, 89, 54, 61, 83, 92, 87, 78, 65, 63, 85, 81, 73

- a. Draw a back to back stem and leaf plot.

[4 marks]



	Range	Median	Mode	Mean
Marcus's team	64	67	56	71.1
Roger's team	38	78	no mode	76

- b. They also calculated the table above. Which team do you believe is more successful?

[1 marks]

Roger - Higher mean/median

- c. Roger's team plays 1 extra game and only scores 34 points. Describe how this will affect

[3 marks]

- i. The mean.

Reduces it to 73.375

- ii. The mode

No effect

- iii. The median

Reduces it ~~to~~ to 76